RAILWAY NETWORK SIMULATION SYSTEM USER MANUAL

Integrative Project – Second Semester

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Introduction

Welcome to the User Manual for the Railway Network Simulation System, developed as part of the Integrative Project. This document is designed to guide users through the system's main features and functionalities, ensuring they can effectively navigate and utilize the software developed throughout the semester.

Purpose and Scope

The objective of this manual is to provide clear, accessible instructions for using the railway network simulation platform. It covers both the map and scenario editor and the simulation environment, offering users the tools to design, manage, and analyse virtual railway systems. Whether the user is building a fictional city network or testing logistical strategies, this guide ensures a smooth and informed experience with the platform.

The product described in this manual is an educational software solution inspired by classic simulation games like Railroad Tycoon. It aims to simulate the operation of railway networks in a simplified yet insightful way. Users can create maps with cities and industries, define historical and technological scenarios, and manage stations, trains, and routes. Additionally, it includes tools for statistical analysis and topological validation, allowing users to assess the performance and connectivity of the network.

Target Audience

This manual is intended for two types of users:

- **Editors**: Users who create and configure maps and scenarios. This role involves defining the geographical and temporal context of the simulation.
- **Players**: Users who engage with the simulation, making strategic decisions such as building stations, buying locomotives, and setting train routes. In future iterations, the platform may support multiplayer experiences.

Whether you're a student building the project, a professor reviewing its progress, or a user testing the final product, this manual aims to offer a comprehensive and user-friendly overview of the platform. Let's begin your journey into the dynamic world of railway management.